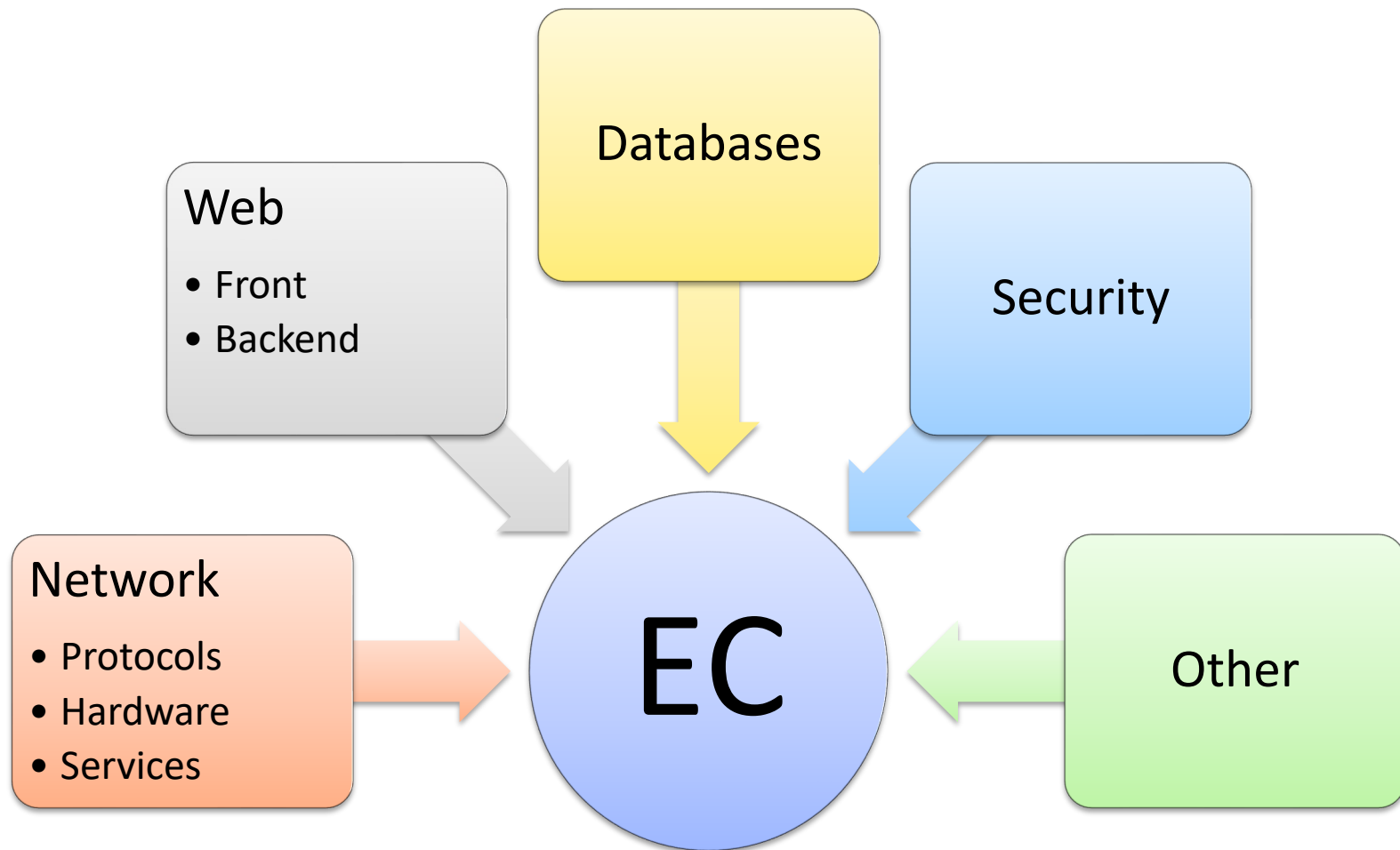




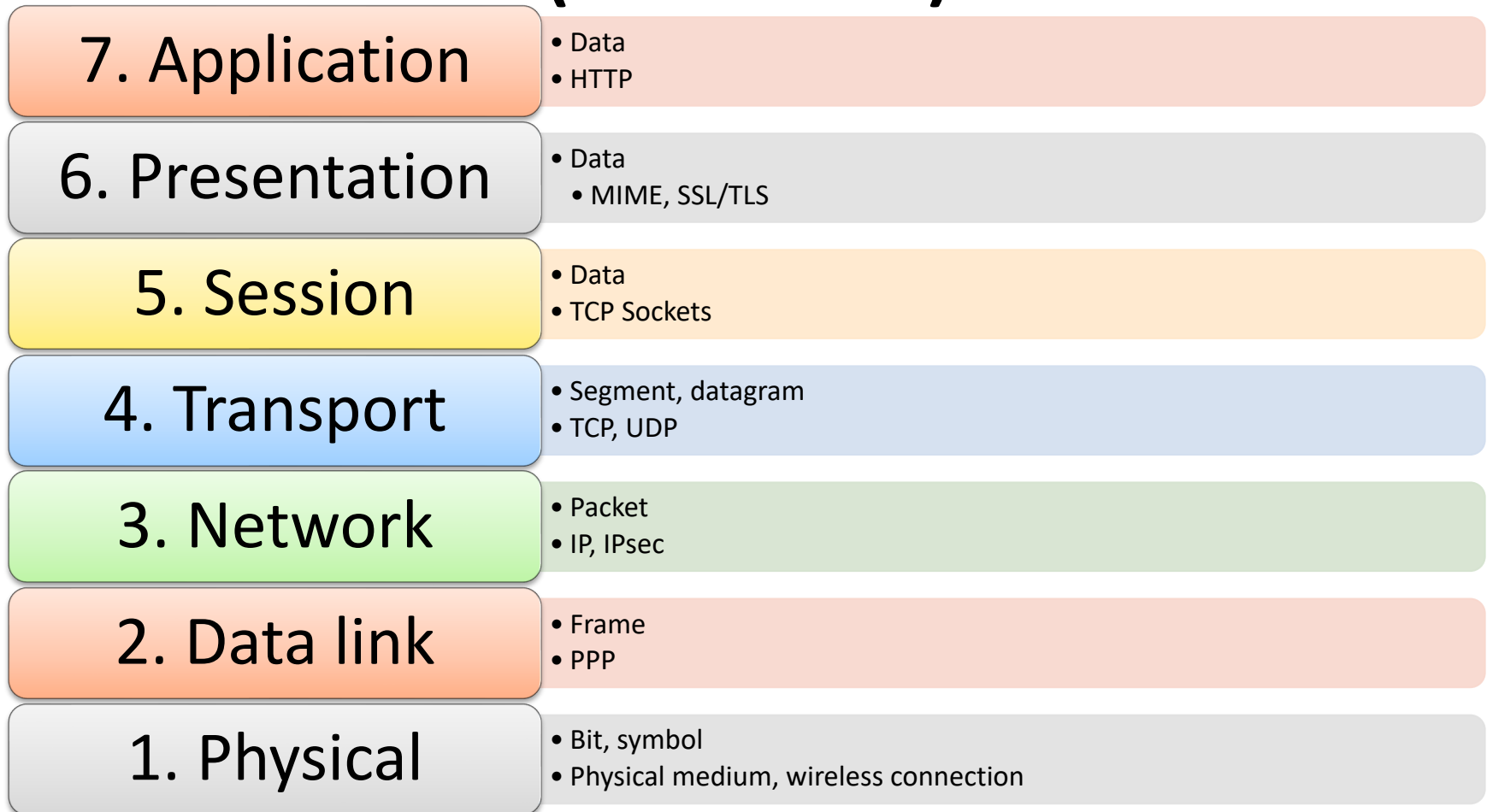
WIESŁAW PIETRUSZKIEWICZ

TECHNOLOGICAL STACK

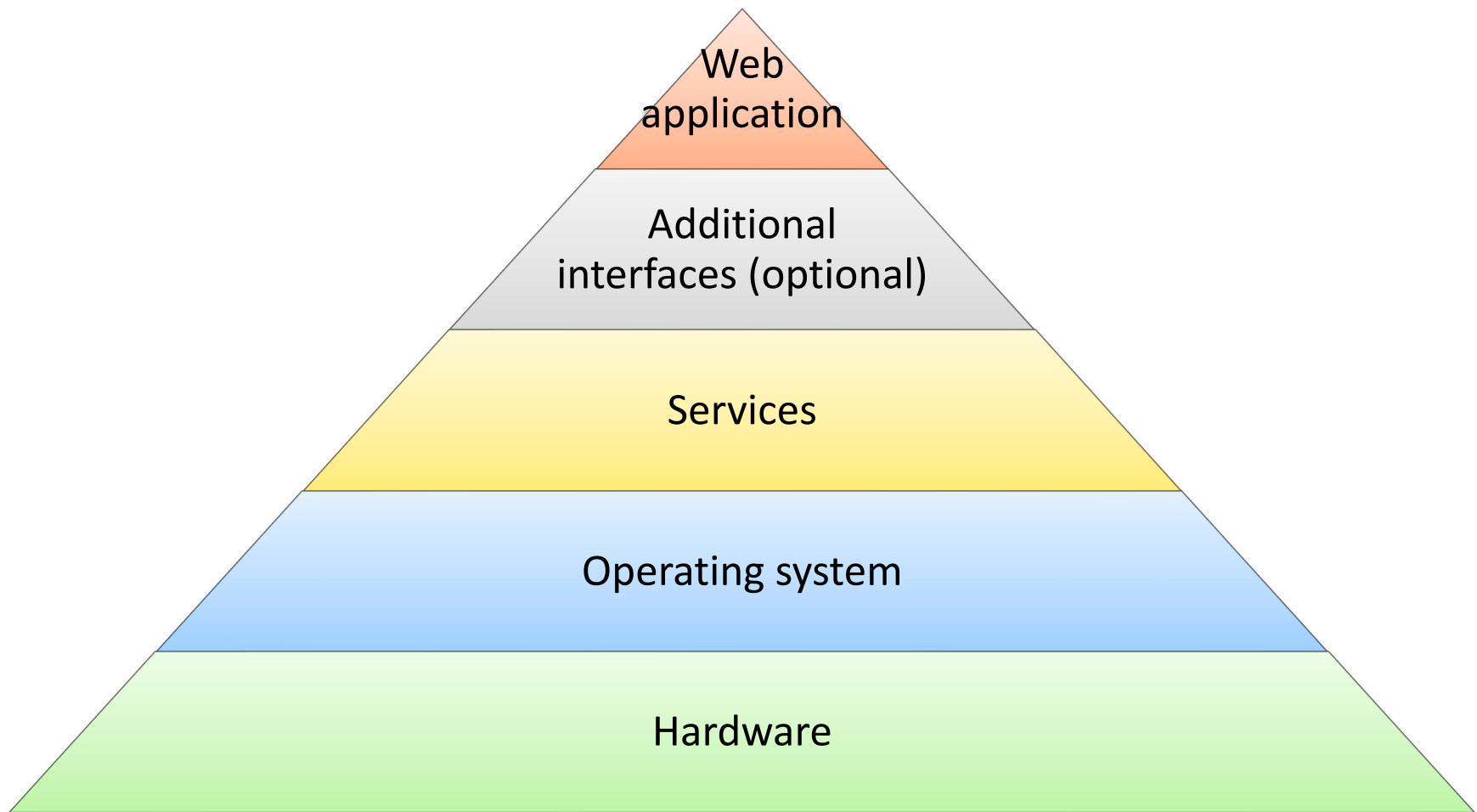
IT-foundation of E-commerce



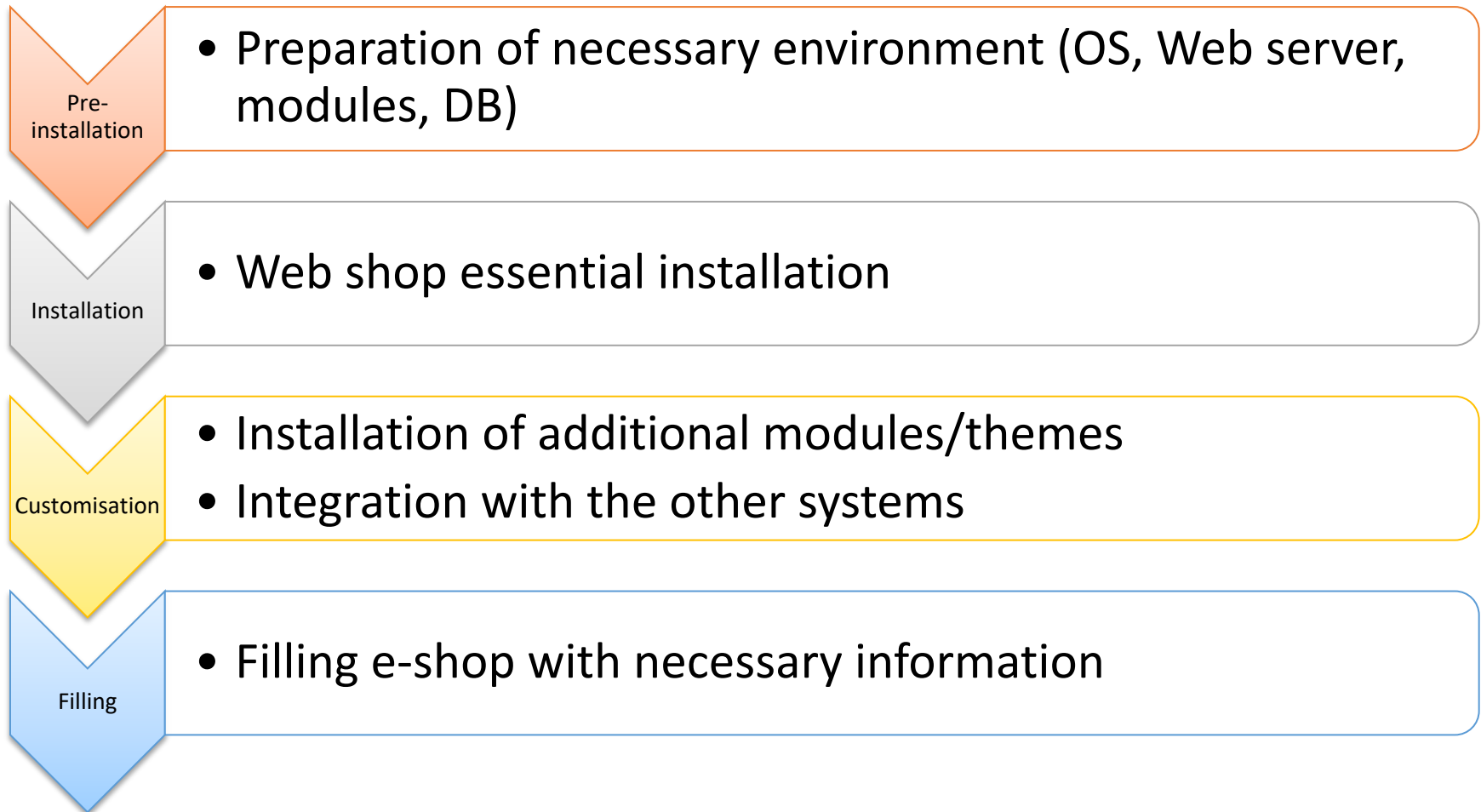
The Open Systems Interconnection model (OSI Model)



Typical architecture of a web site/app



E-shop – Deployment



E-shop core IT requirements

- Typical requirements to deploy and run e-shop include:
 - Hosting space
 - Accessibility via domain
 - Database service (MySQL/MariaDB)
 - Webserver with a support of dynamic languages (Apache & PHP)

Typical server configuration

LAMP	WAMP (development > production)	MEAN (alternative)
• Linux • Apache • MySQL/MariaDB • PHP/Perl/Python	• Windows • Apache • MySQL/MariaDB • PHP/Perl/Python	• MongoDB • Express.js • Angular • Node.js

LAMP

Use Cases

- LAMP is commonly used for developing & running PHP-based applications, including CMSs, Web carts.
- It is suitable for a wide range of web applications, from small personal websites to large enterprise solutions.

Advantages

- Open-source and widely supported, with a large community and extensive documentation.
- Well-suited for traditional web applications that rely on relational databases.
- Easy to set up and configure on Linux servers.

Disadvantages

- Limited to relational database structures, which may not be ideal for applications requiring NoSQL databases.
- PHP may not be as performant as some other server-side languages in certain scenarios.
- Lack of sophisticated libraries for PHP

WAMP

Use Cases

- WAMP is commonly used for developing PHP-based applications, including CMSs, Web carts.
- It is suitable for local development environments, allowing developers to test their applications on their Windows machines before deploying them to a live server.

Advantages

- Easy to set up and configure on Windows.
- Provides a familiar environment for developers who are accustomed to Windows.
- Supports a wide range of PHP frameworks and libraries.

Disadvantages

- All for given for LAMP
- Limited to Windows operating systems, which may not be ideal for all production environments.

MEAN

Use Cases

- MEAN is ideal for building modern web applications, particularly single-page applications (SPAs) that require real-time data updates and a dynamic user experience.
- It is commonly used in applications that require a fast, scalable, and responsive architecture (event-driven).

Advantages

- Unified full-stack JavaScript development environment.
- MongoDB's structure provides flexibility in data modelling and scalability.
- Real-time capabilities with Node.js - suitable for apps requiring instant data updates (e.g., chat applications, collaborative tools).

Disadvantages

- The learning curve may be steeper for developers unfamiliar with JavaScript or the MEAN stack components.
- Requires a good understanding of asynchronous programming, which is a key feature of Node.js.

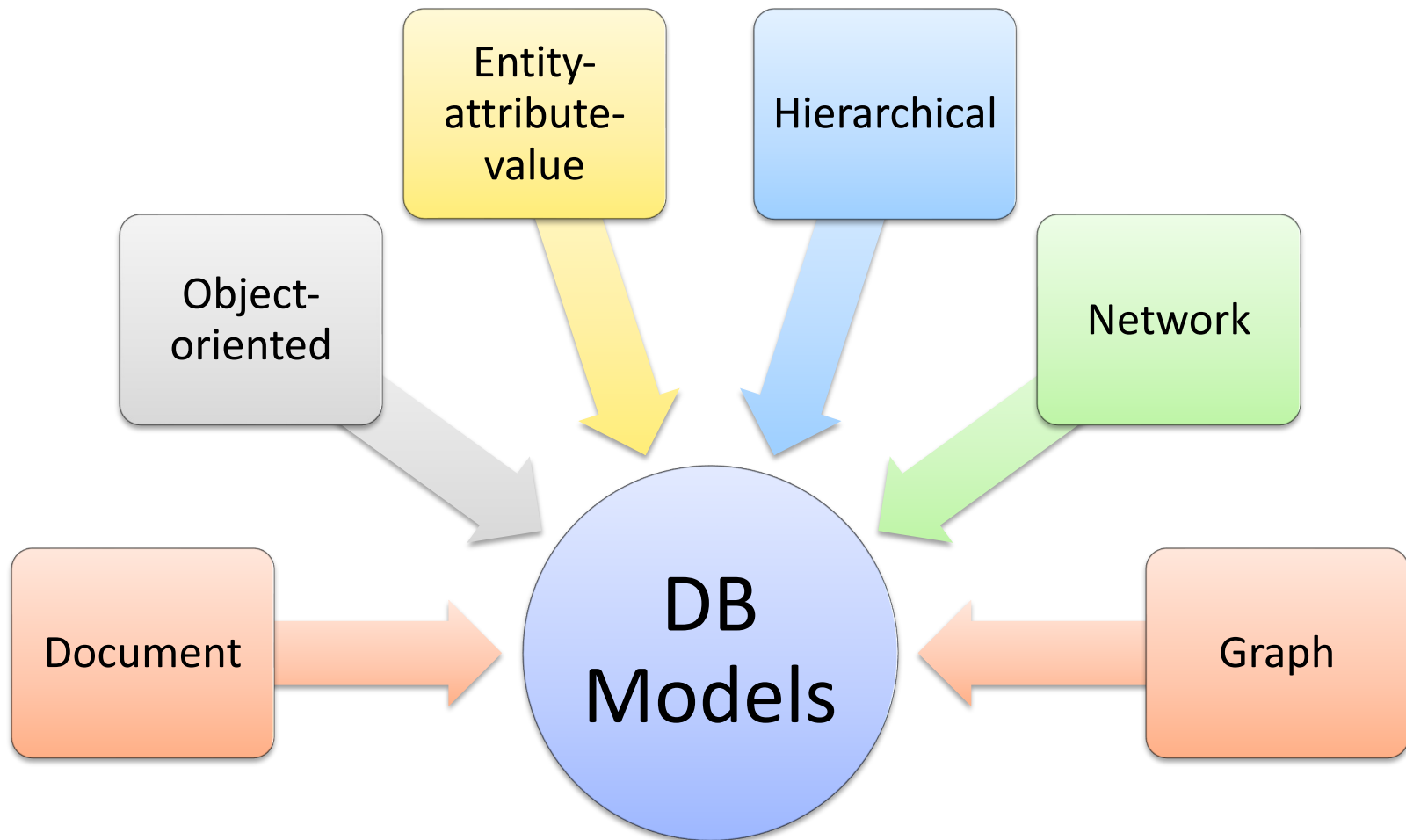
Database in Web Systems

- Database (service/server) is a key element commonly used by web systems to store information
- Used for the information storage, as well as for the information processing
 - Stores pieces of web pages, to be later recreated & delivered as the content (website)
 - Stores information necessary to provide functionality (webapps)

Database - Relational

- The most common model of DB build over SQL
- Assumes the stability of structure and relations between elements
- A part of common hosting packages
- A safe and good choice
- Free RDBes:
 - MySQL
 - MariaDB
 - PostgreSQL
 - SQLite (embedded, not a service)
- Commercial RDBes:
 - Oracle Database
 - Microsoft SQL Server
 - IBM Db2

Other DB Models for specific EC domains



CONNECTION

Bandwidth

- **Bandwidth** refers to the maximum rate at which data can be transmitted over a network connection in a given amount of time, typically measured in bits per second (bps). It represents the capacity of the network link.
- **Throughput** Throughput is the actual rate at which data is successfully transmitted over a network in a given amount of time, also measured in bits per second (bps). It reflects the real-world performance of the network.

Bandwidth

- **Goodput** refers to the amount of useful data transmitted over a network in a given amount of time, excluding all protocol overhead, retransmissions, and any other non-useful data. It is also measured in bits per second (bps).

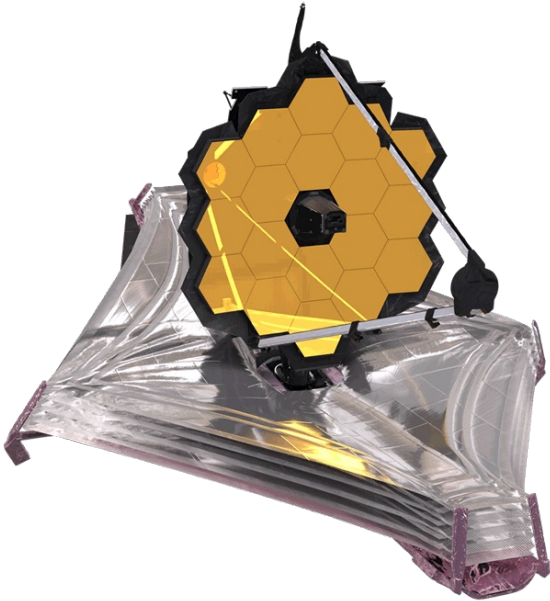
$$\text{Goodput} < \text{Throughput} < \text{Bandwidth}$$

- Speed (of any kind) is usually a key value provided for every connection
- Could be symmetrical or asymmetrical

Bandwidth comparison

- | | | | |
|-----------------|------------------|--------------|--|
| • 56 kbit/s | Modem / Dialup | • 622 Mbit/s | OC12 |
| • 384 kbit/s | WCDMA (UMTS) | • 1 Gbit/s | Gigabit Ethernet |
| • 1.5 Mbit/s | ADSL Lite | • 1 Gbit/s | LTE Advanced (LTE+) |
| • 1.544 Mbit/s | T1/DS1 | • 1.3 Gbit/s | Wireless 802.11ac |
| • 2.048 Mbit/s | E1 / E-carrier | • 2.5 Gbit/s | OC48 |
| • 4 Mbit/s | ADSL1 | • 4 Gbit/s | 5G (max for mmWaves) |
| • 10 Mbit/s | Ethernet | • 5 Gbit/s | SuperSpeed USB |
| • 11 Mbit/s | Wireless 802.11b | • 7 Gbit/s | Wireless 802.11ad |
| • 24 Mbit/s | ADSL2+ | • 9.6 Gbit/s | OC192 |
| • 28 Mbit/s | HSPA+ | • 10 Gbit/s | 10 Gigabit Ethernet,
SuperSpeed USB 10 Gbit/s |
| • 44.736 Mbit/s | T3/DS3 | • 20 Gbit/s | SuperSpeed USB 20
Gbit/s |
| • 54 Mbit/s | Wireless 802.11g | • 40 Gbit/s | Thunderbolt 3 |
| • 100 Mbit/s | Fast Ethernet | • 100 Gbit/s | 100 Gigabit Ethernet |
| • 155 Mbit/s | OC3 | | |
| • 300 Mbit/s | LTE | | |
| • 600 Mbit/s | Wireless 802.11n | | |

Connection asymmetry – Example



- S-band, telemetry and telecommand:
 - Up 16 kbit/s
 - Down 40 kbit/s
- K_a-band, science data downlink:
 - Down: max 28 Mbit/s
- For web-based solutions:

$$\text{Data}_{\text{Client} \rightarrow \text{Server}} < \text{Data}_{\text{Server} \rightarrow \text{Client}}$$

Latency

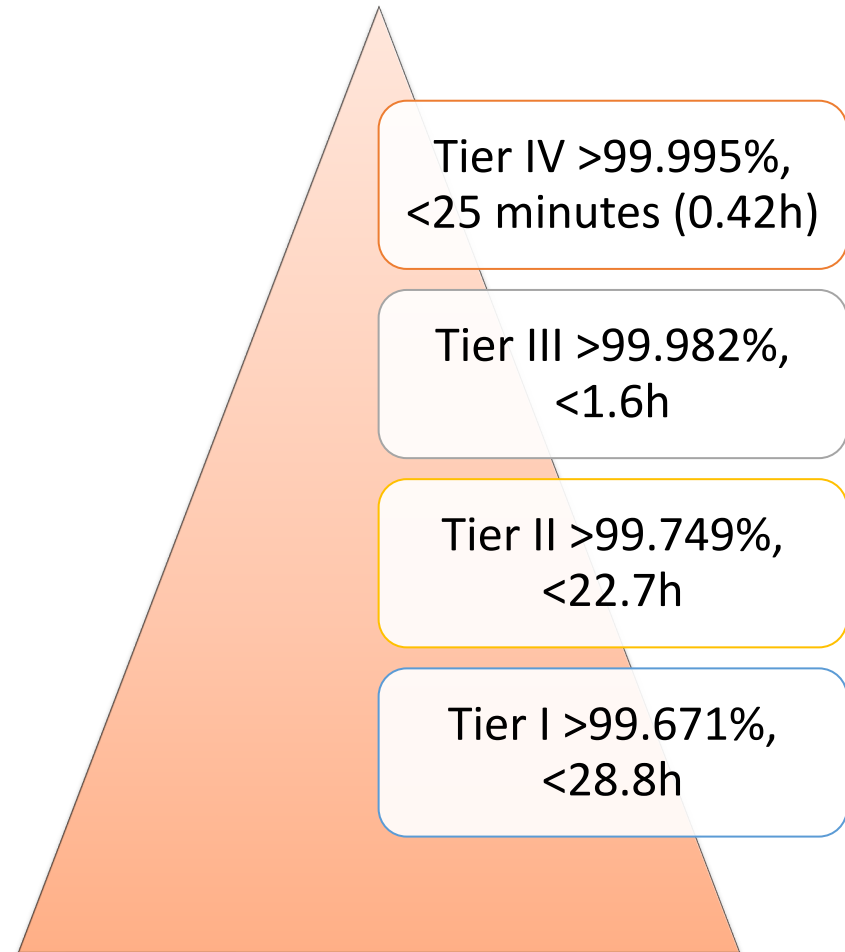
- The latency is a time delay measuring how long it takes data to travel between the sender and receiver nodes
- An essential parameter for interactive remote environments (expected low values)
- Essential for interactive solutions
- Less important (up to a certain degree) for web sites

Transfer

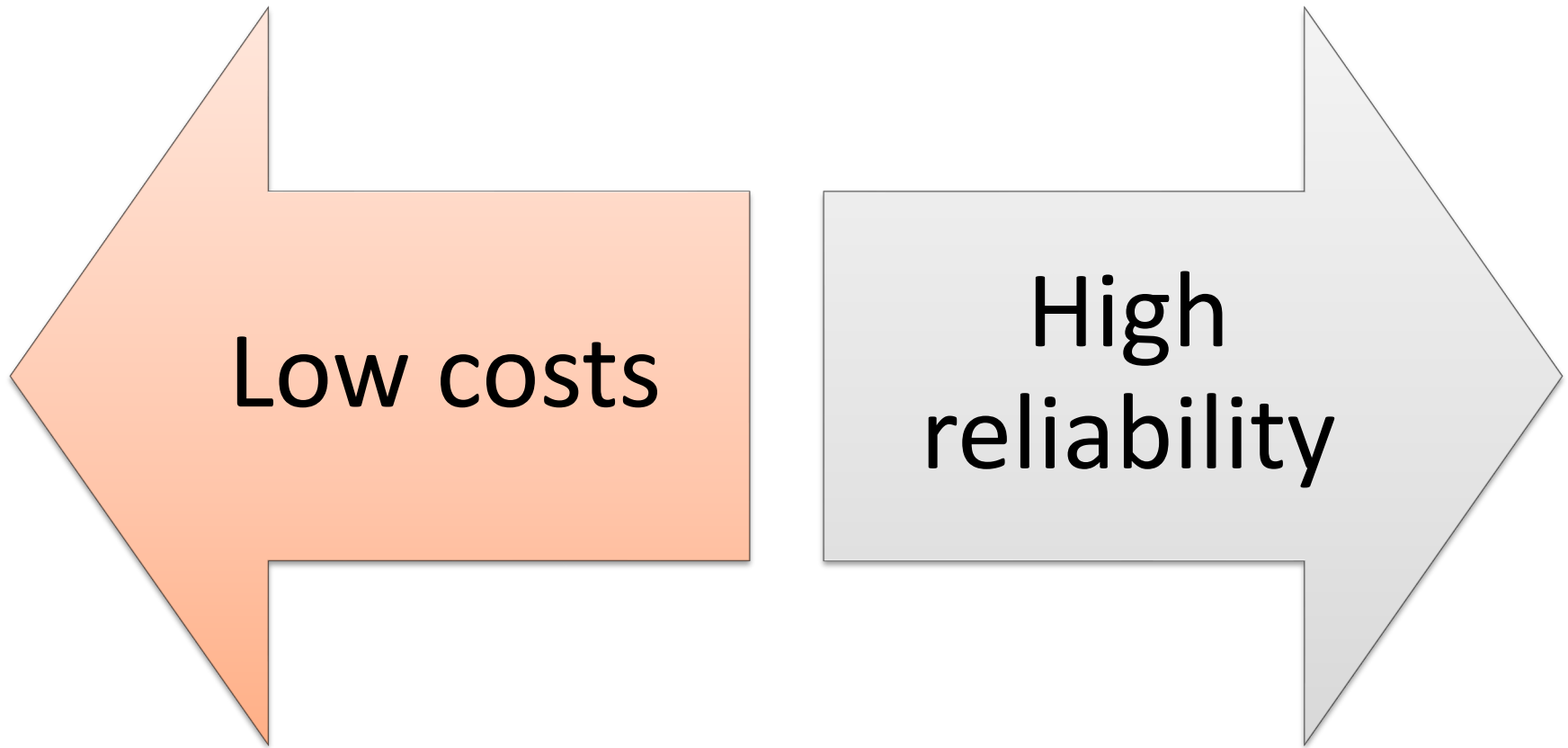
- Transfer denotes an amount of data that will/can be/was transferred
- Could be measured at different levels (OSI layers), usually the most profitable to a service provider
- Possible transfer limits – hard (transfer interrupted) or soft (bandwidth reduction)
- Limits often found for cheapest services

Reliability #1 - Data centre reliability levels by Uptime Institute

- Reliability - a parameter representing availability of machine/es
- Downtime – measured in hours per year
- Availability – operability relative to the total operating time



Reliability - Choice



Reliability – Issues & Countermeasures

Reliability issues

- Failures (unplanned)
- Maintenance (planned)
- Additional factors (including human-based deliberate actions)
- Unavoidable risks

Countermeasures

- Uninterruptible power supplies
- Power generators
- Protected access to the equipment
- Prepared facilities
- Redundant hardware
- Redundant connections
- Backups
- Training and control of staff

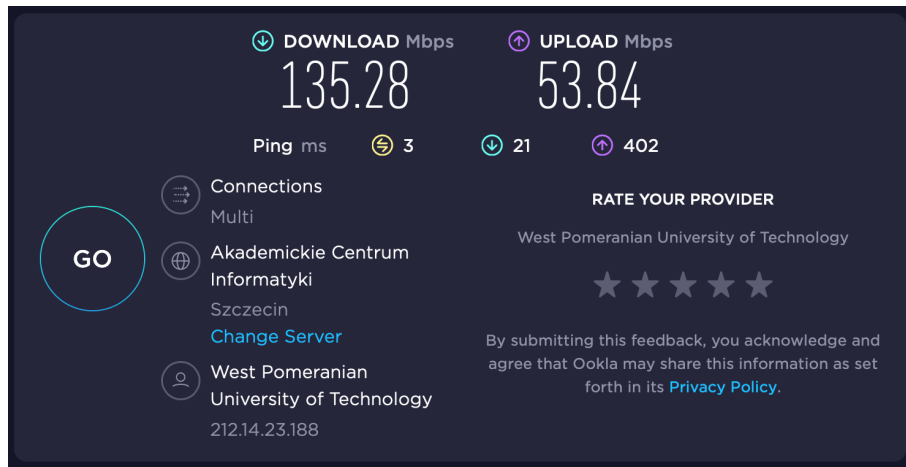
Stability of connection parameters

- Many solutions offer automatic adjustment of content (effecting the required bandwidth)
- It is estimated by test runs
- Stability of connection causes estimates to be less or more accurate
- Overestimation vs underestimation of connection parameters is different from user's perspective

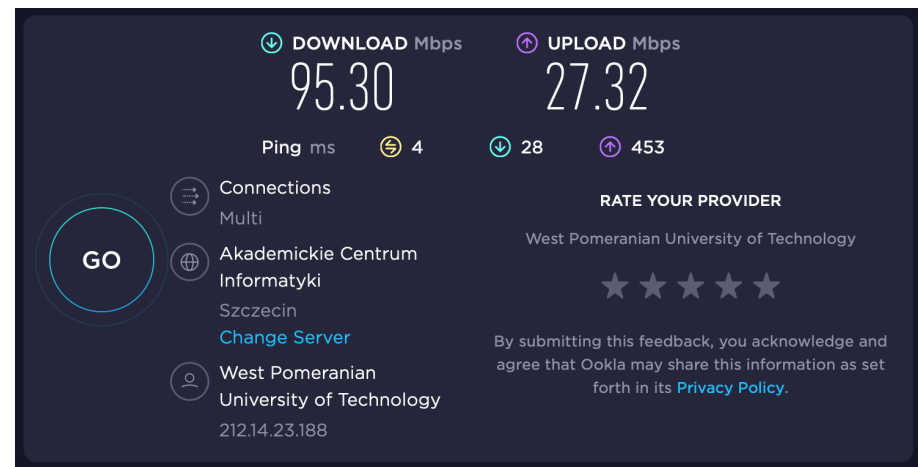
Speed measurements

Faculty WiFi – Example done by www.speedtest.net

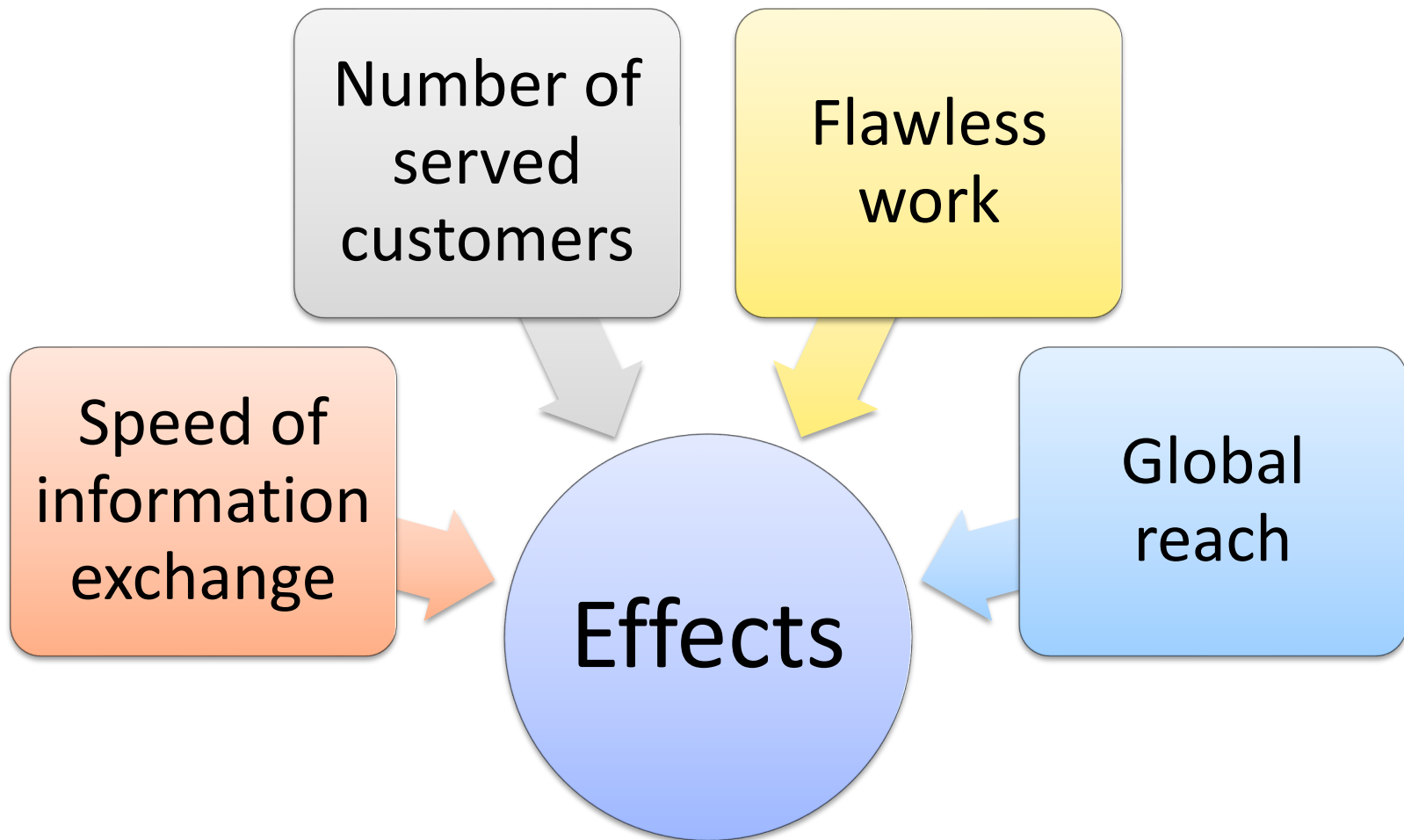
8:59



9:01

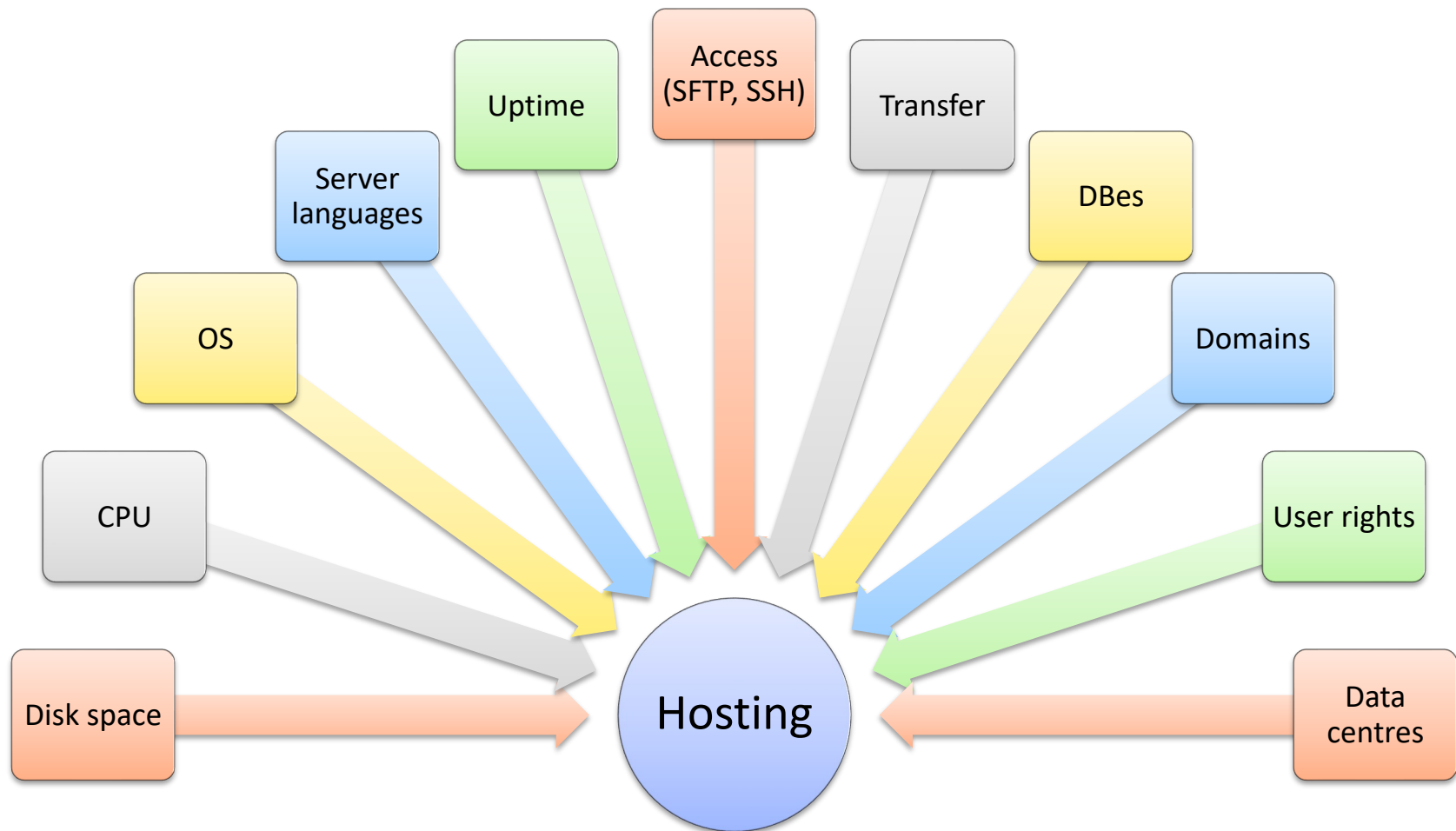


Implications of connection for EC



HOSTING

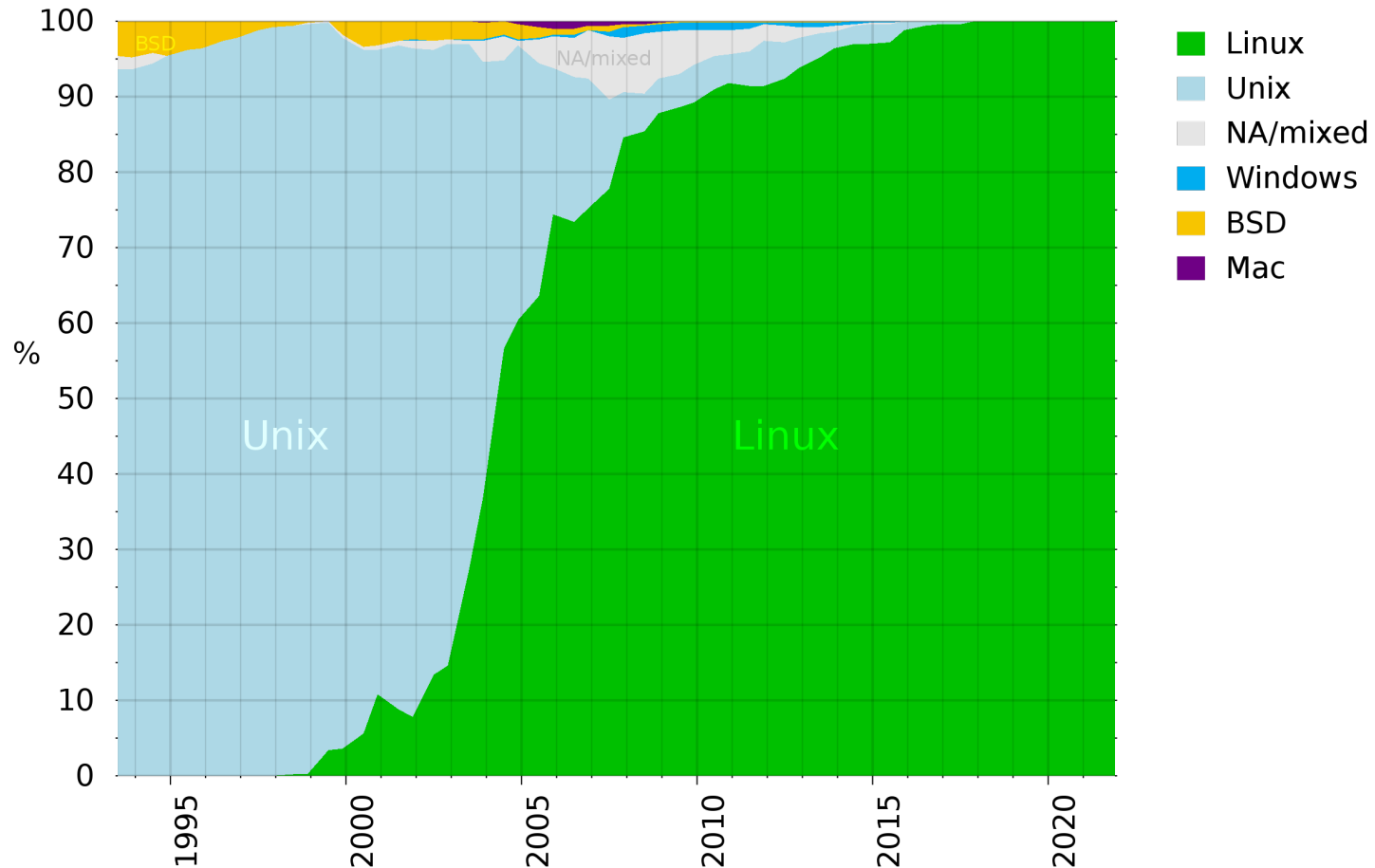
Hosting – Main features



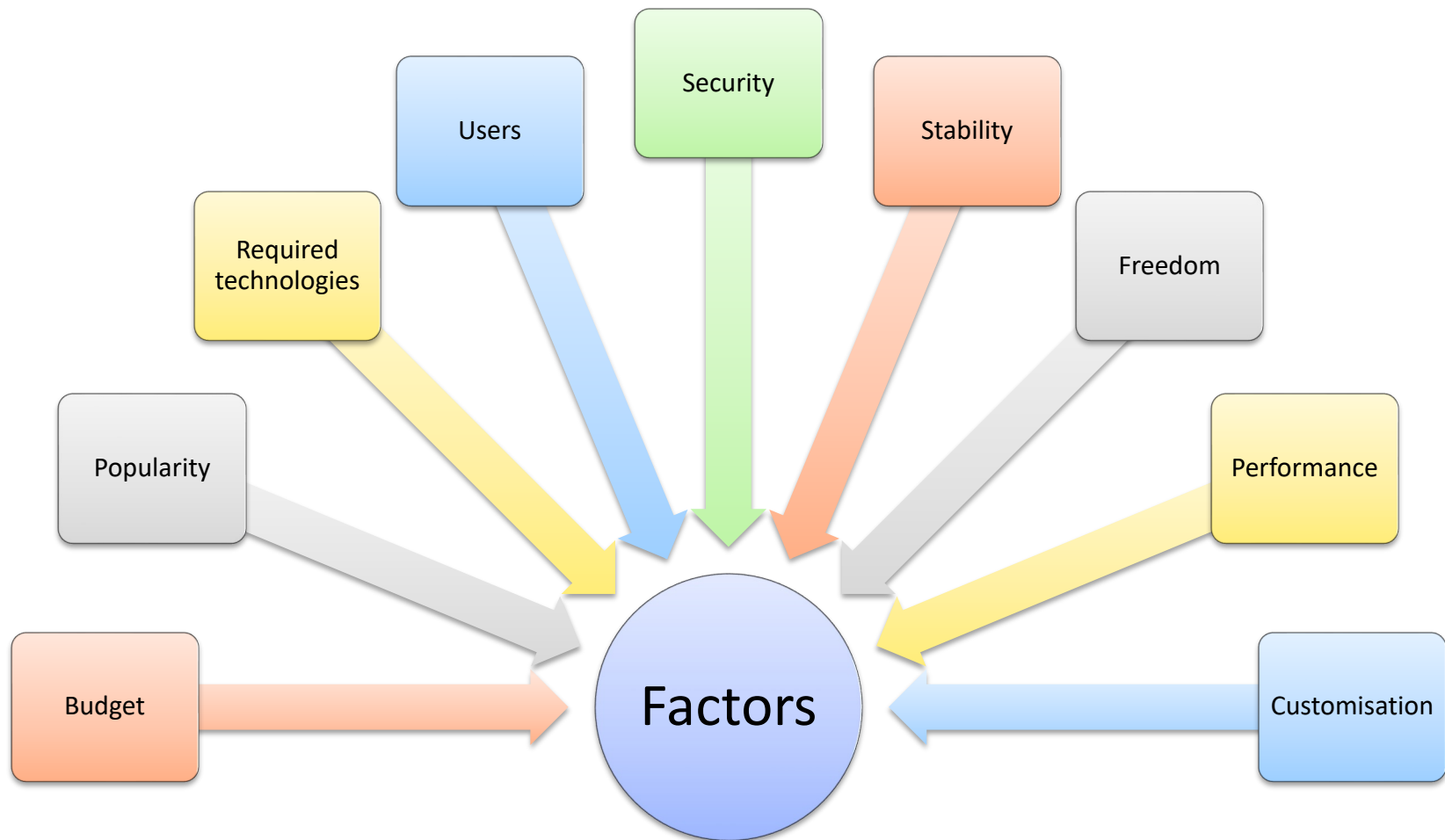
Hosting – Operating Systems

- Operating system is a software foundation of every computer system
- Popularity of server OSes (W3Tech – November 2024):
 - “UNIX” (including Linux) 87.8%
 - Windows 12.2%
- Server OS differs from the desktop OS in:
 - Features
 - Stability
 - Performance
 - Customisation
 - Security
 - Addons

OS – Supercomputers (Source: Benedikt.Seidl)



OS – Selection factors



Linux distributions (distros)

- Linux distribution is a software collection of:
 - kernel, init system, GNU tools&libraries, manuals;
 - other tools, display server, desktop environment, user apps;
 - package management system.
- Approx. 1000 distros exist, while only a small fraction is popular.
- Distribution can be based on another one.

Linux distros

- Distros differs in:
 - Character (commercial/personal);
 - Device-orientation (server/desktop/embedded devices);
 - Main quality (usability, easiness of installation / operation, security, portability, performance);
 - Main purpose (general environment/firewalls, routers, network nodes, virtualisation);
 - Supported hardware.

Distros – popularity distrowatch.com

Last 12 months			Last 6 months			Last 3 months			Last 1 month		
1	MX Linux	2712↔	1	MX Linux	2625▲	1	MX Linux	2473▼	1	MX Linux	2715▲
2	Mint	2099▲	2	Mint	2131▼	2	Mint	2134▲	2	Mint	2029▲
3	EndeavourOS	2041▼	3	EndeavourOS	1729▼	3	EndeavourOS	1561▼	3	EndeavourOS	1609▲
4	Manjaro	1377▼	4	Debian	1538▲	4	Debian	1478↔	4	Debian	1541↔
5	Debian	1322▲	5	Manjaro	1212▼	5	Manjaro	1200↔	5	Ubuntu	1299▲
6	Ubuntu	1082↔	6	Ubuntu	1023↔	6	Ubuntu	1132▲	6	Manjaro	1137▲
7	Pop!_OS	1058▼	7	Pop!_OS	912↔	7	Fedora	870▲	7	Fedora	980▲
8	Fedora	1053↔	8	Fedora	873▼	8	Pop!_OS	836▼	8	Pop!_OS	862▲
9	openSUSE	749↔	9	openSUSE	782↔	9	openSUSE	718▲	9	Garuda	792▲
10	Lite	673▼	10	Zorin	640▲	10	Zorin	607↔	10	openSUSE	767↔
11	Zorin	641↔	11	Lite	588↔	11	Garuda	605▲	11	Zorin	688▲
12	Garuda	624▲	12	Garuda	573▲	12	antiX	558↔	12	elementary	539▼
13	elementary	558↔	13	antiX	505↔	13	Lite	558▼	13	FreeBSD	532▲
14	KDE neon	545↔	14	KDE neon	500↔	14	elementary	527▲	14	Nobara	523▼
15	antiX	494↔	15	elementary	492▲	15	Mageia	520↔	15	KDE neon	512▲
16	Kali	410↔	16	Mageia	472↔	16	Nobara	465▲	16	antiX	487▲
17	PCLinuxOS	408↔	17	Nobara	464▲	17	KDE neon	459▲	17	Kubuntu	450▲
18	Puppy	404↔	18	Vanilla	394▲	18	FreeBSD	407▲	18	Lite	442▲
19	Mageia	403▲	19	NixOS	376▲	19	Kali	398↔	19	AlmaLinux	439▲
20	AlmaLinux	397↔	20	PCLinuxOS	375↔	20	Vanilla	376▲	20	Vanilla	434▲

Hosting-types

Shared

- Shared resources
- Limited user rights
- Limited functionality
- No customisation
- No maintenance required
- Cheap

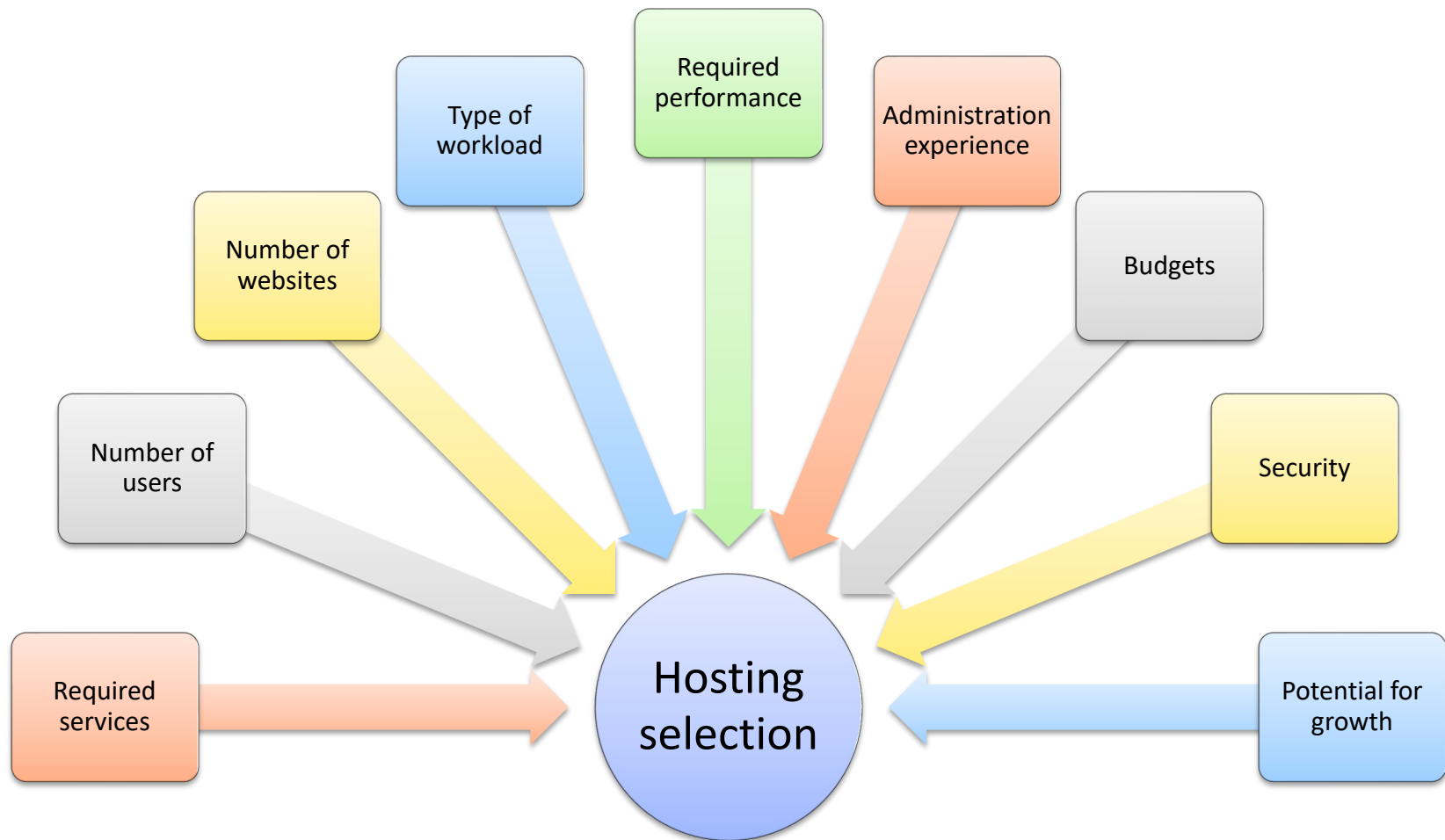
Dedicated

- Full control
- Availability of resources
- Customised
- High costs
- Maintenance required

Virtual

- Full control
- Availability of assigned virtual resources
- Customised
- Average costs
- Scalable
- Maintenance required

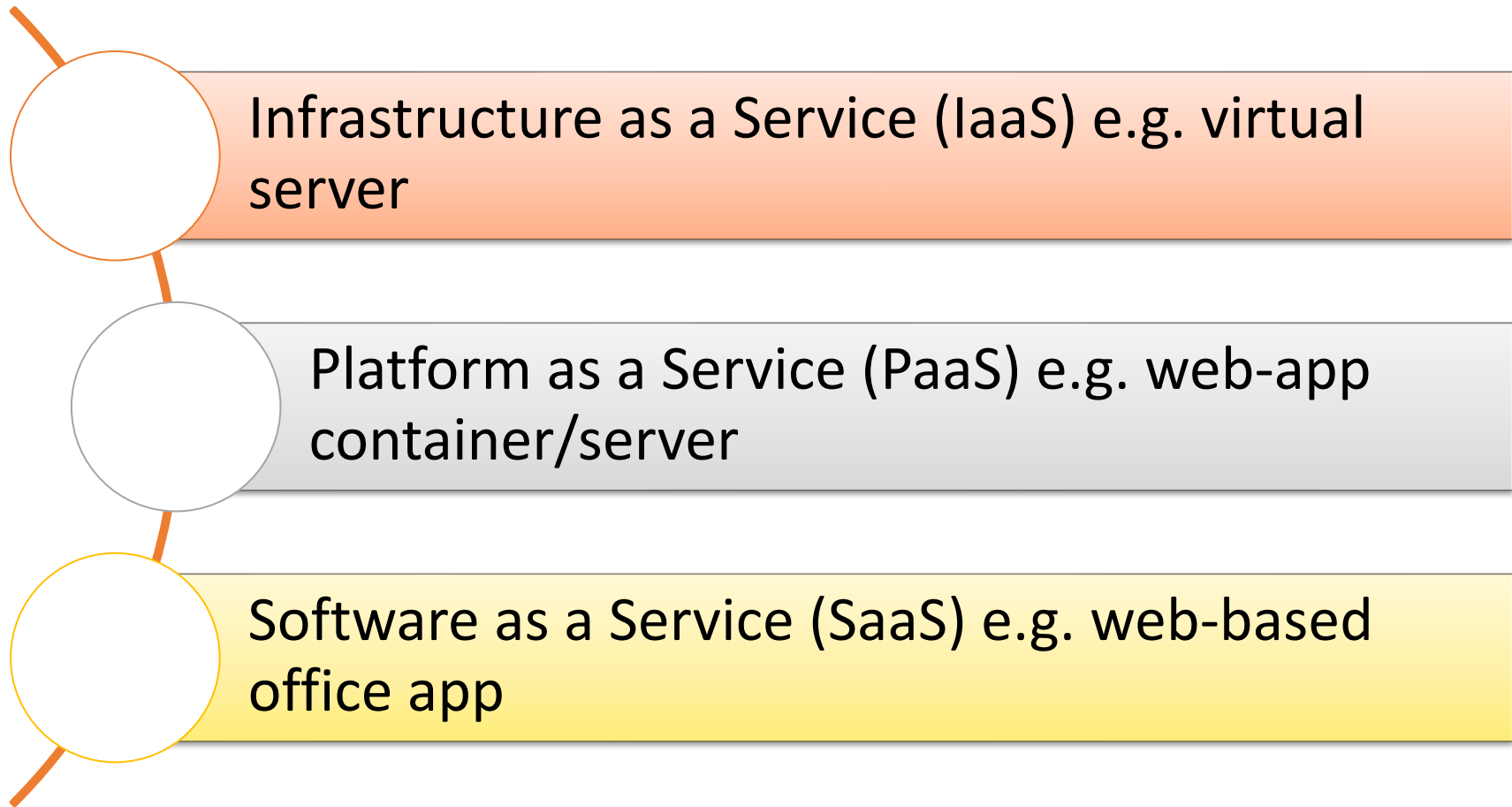
Hosting-type selection factors



Hosting location – AWS Global Infrastructure Map (November 2022)

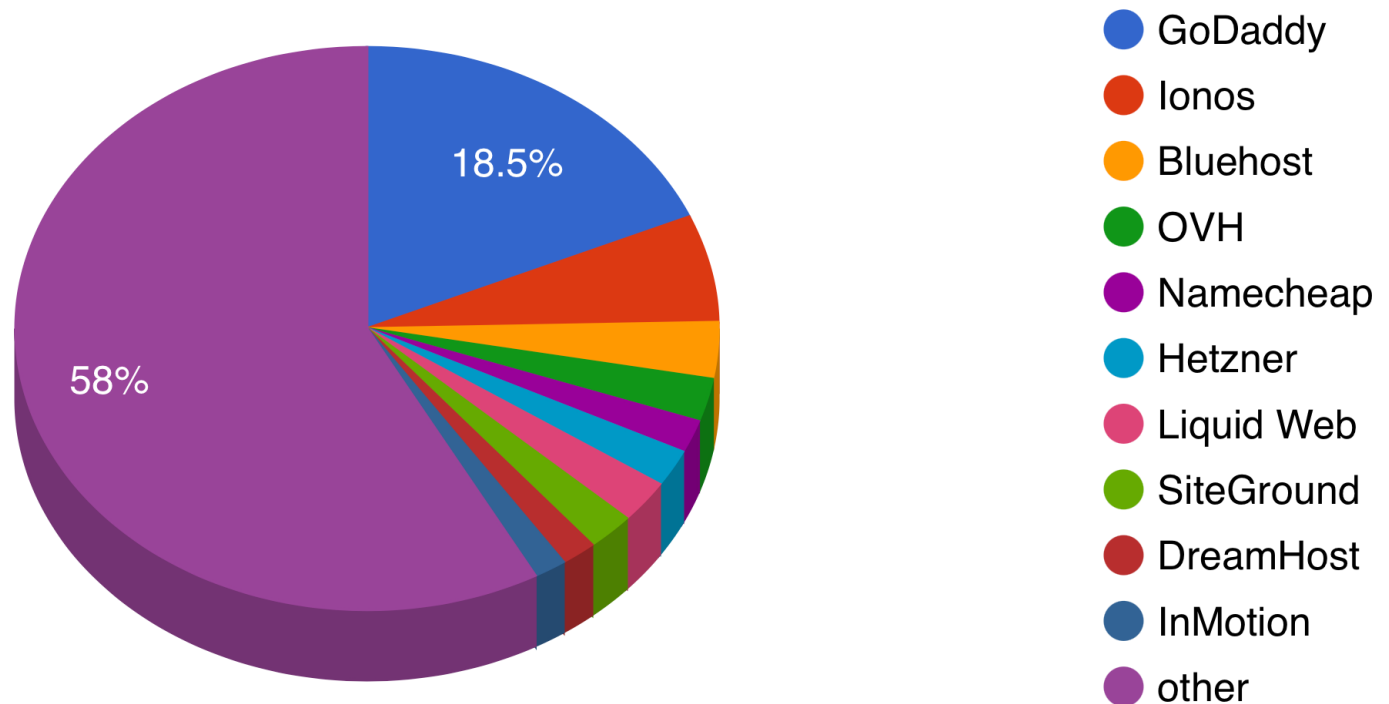


Cloud services



Popular global hosting (October 2023)

Market size of the largest web hosts



Source:codeinwp.com

GoDaddy

- US-based #1 hosting company
- Data centres – North America, Europe, Asia
- .com €1/€20
- Hosting from €3/month

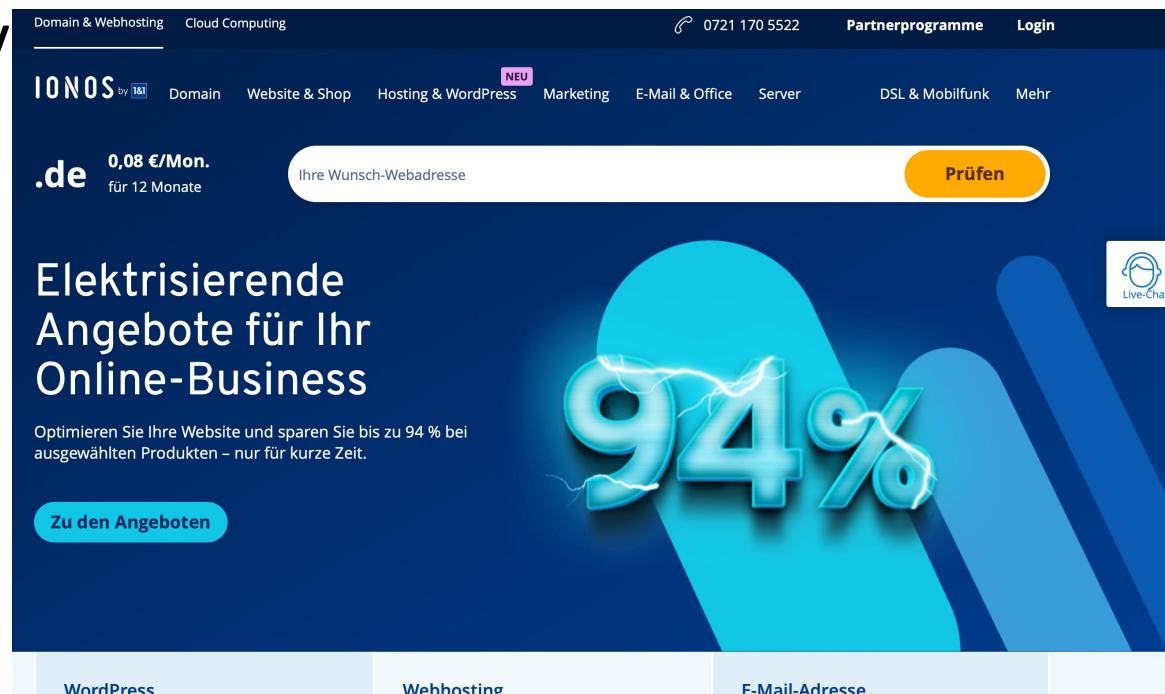
The screenshot shows the GoDaddy website homepage. At the top, there's a navigation bar with the GoDaddy logo and links for Domain Names, Websites & Stores, Website Security, and Marketing Tools. A search bar prompts users to 'Find your perfect domain' with a 'Search Domain' button. Below this, a large banner for 'Websites + Marketing' features a woman in an orange shirt and the text 'Create your website.' with 'Start for Free' and 'Learn more' buttons. To the right, a brown box promotes 'Domain Names' with the offer 'Grab a .com for €0.99*/1st yr.' and a 'Find Your Domain' button. At the bottom, a table lists various services and their prices:

.com €0.99*/1st yr	Web Hosting €2.99/mo	WordPress €3.99/mo
Professional Email €2.49 per user/mo	SSL €55.99/yr	VPS Hosting €4.99/mo

On the far right, a purple box promotes '.CLUB' with the offer 'Sale! €0.85*/1st yr. Bring everyone together online.' and a 'Contact Us' button.

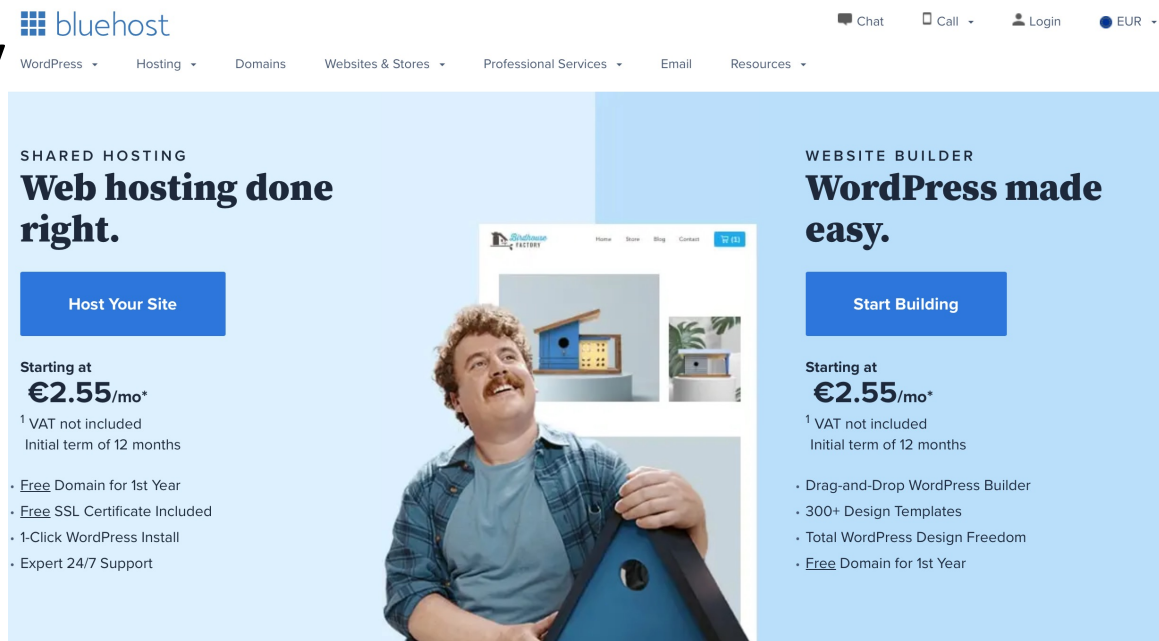
Ionos

- German company
- #2 hosting company
- Data centres – Europe & US
- .com € 0.7(2)/mth
- Hosting € 2(4)/mth



Bluehost

- US-based company
- #3 hosting company
- Data centres – US, UK, India, Hong Kong, China
- .com € 11.25/year
- Hosting € 2.55/mth

The image shows the Bluehost website homepage. At the top, there is a navigation bar with the Bluehost logo and links for WordPress, Hosting, Domains, Websites & Stores, Professional Services, Email, and Resources. On the right side of the navigation bar, there are links for Chat, Call, Login, and a currency selector set to EUR. The main content area is divided into two columns. The left column is titled 'SHARED HOSTING' and features the headline 'Web hosting done right.' Below this is a blue button labeled 'Host Your Site'. Further down, it states 'Starting at €2.55/mo*' with a footnote '1 VAT not included Initial term of 12 months'. A list of features includes 'Free Domain for 1st Year', 'Free SSL Certificate Included', '1-Click WordPress Install', and 'Expert 24/7 Support'. The right column is titled 'WEBSITE BUILDER' and features the headline 'WordPress made easy.' Below this is a blue button labeled 'Start Building'. Further down, it states 'Starting at €2.55/mo*' with a footnote '1 VAT not included Initial term of 12 months'. A list of features includes 'Drag-and-Drop WordPress Builder', '300+ Design Templates', 'Total WordPress Design Freedom', and 'Free Domain for 1st Year'. In the center of the page, there is a man with a mustache, wearing a blue shirt, holding a large blue folder or tablet. The background of the page is light blue with white text and blue buttons.

BLUEHOST WEBSITE BUILDER

Hosting analysis (Ionos) – Shared

The image displays four hosting plans from Ionos: Essential, Business, Pro, and Premium. The Pro plan is highlighted as the 'Meistgekauft' (Most bought) option. Each plan includes details on duration, pricing (initial and subsequent months), setup fees, and a list of features with information icons.

Plan	Duration	Price (€/Mon.)	Setup Fee (€)
Essential	6 Monate	2,–	15,–
Business	6 Monate	4,–	15,–
Pro	12 Monate	1,–	10,–
Premium	6 Monate	10,–	15,–

Feature	Essential	Business	Pro	Premium
Webpace	50 GB	100 GB	250 GB	500 GB
Datenbanken	5 MariaDB oder MySQL	100 MariaDB oder MySQL	250 MariaDB oder MySQL	500 MariaDB oder MySQL
Domains	1 inklusive	2 inklusive	3 inklusive	4 inklusive
E-Mail-Adresse	Professionelle E-Mail-Adresse inklusive	Professionelle E-Mail-Adresse inklusive	Professionelle E-Mail-Adresse inklusive	Professionelle E-Mail-Adresse inklusive
Leistung	-	Grundrechenleistung	Doppelte Leistung	Dreifache Leistung
CDN	-	-	-	Content Delivery Network (CDN)

Hosting analysis (Ionos) – Shared #2

Fast Loading Times				
Performance Level	optional	Initial performance level 1	Initial performance level 2	Initial performance level 3
RAM	Up to 2.5 GB RAM (1.25 GB always guaranteed)	Up to 2.5 GB RAM (1.25 GB always guaranteed)	Up to 6 GB RAM (3 GB always guaranteed)	Up to 9 GB RAM (4.5 GB always guaranteed)
PHP memory limit	512 MB	512 MB	640 MB	640 MB
PHP with OPcache ⓘ	✓	✓	✓	✓
CDN with Railgun™ ⓘ	Optional	Optional	Optional	✓
HTTP/2	✓	✓	✓	✓
gzip	✓	✓	✓	✓

Hosting analysis (Ionos) – Shared #3

Apps				
Easy 1-Click installation for over 70 applications	✓	✓	✓	✓
WordPress (Performance)				
Joomla! (Performance)				
Drupal (Performance)				
TYPO3 (Performance)				
PrestaShop (Performance)				
WebApps	✓	✓	✓	✓

Hosting analysis (Ionos) – Shared #4

Webspace				
Webspace	50 GB	100 GB	250 GB	Unlimited
Inode file limit ⓘ	262,144	262,144	262,144	262,144
Unlimited Traffic ⓘ	✓	✓	✓	✓
Unlimited (S)FTP Accounts	✓	✓	✓	✓
Webspace Explorer	✓	✓	✓	✓
Protected Directories (.htaccess)	✓	✓	✓	✓
Version Control Management (git)	✓	✓	✓	✓
Shell Access (SSH)	✓	✓	✓	✓
WebDAV	✓	✓	✓	✓
IPv6 Ready	✓	✓	✓	✓

Hosting analysis (Ionos) – Shared #5

Databases and Scripts				
MariaDB 10 or MySQL 5 databases (2 GB each)	5	25	100	Unlimited
PHP 7.4 and 8.1	✓	✓	✓	✓
PHP Extended Support available for PHP 4.0+	✓	✓	✓	✓
Manually changeable PHP version	✓	✓	✓	✓
Zend Framework	✓	✓	✓	✓
Perl, Python, Ruby	✓	✓	✓	✓
SSI (Server Side Includes)	✓	✓	✓	✓
Cron Manager for Cron Jobs	✓	✓	✓	✓

Hosting analysis (Ionos) – Shared #6

Domains				
Free domain for 1 year	1	1	1	1
Included free domain extensions	.co.uk, .uk, .com, .net, .org, .info, .me, .biz, .online	.co.uk, .uk, .com, .net, .org, .info, .me, .biz, .online	.co.uk, .uk, .com, .net, .org, .info, .me, .biz, .online	.co.uk, .uk, .com, .net, .org, .info, .me, .biz, .online
Subdomains	10,000	10,000	10,000	10,000
DNS Management (Domain Name System)	✓	✓	✓	✓
Domain Redirection	✓	✓	✓	✓

Hosting analysis (Ionos) – Dedicated #1

L4i HDD Einstiegs-Hardware für Ihre Webprojekte	L-16 HDD Erweiterte Hardware für anspruchsvolle Webprojekte	XL-32 HDD Leistungsstarke Hardware für umfangreiche Webprojekte	XL-64 HDD Leistungsstarke Hardware für mehrere umfangreiche Webprojekte
dauerhaft nur 27,- €/Mon. monatlich kündbar	3 Monate 40,- €/Mon. danach 60,- €/Mon.	3 Monate 60,- €/Mon. danach 80,- €/Mon.	3 Monate 80,- €/Mon. danach 100,- €/Mon.
Weiter	Weiter	Weiter	Weiter
Intel® Quad-Core	Intel® Xeon® E3-1230 v6	Intel® Xeon® E3-1270 v6	Intel® Xeon® E3-1270 v6
4 Cores (HT) x 3,1 GHz (3,4 GHz Turbo Boost)	4 Cores (HT) x 3,5 GHz (3,9 GHz Turbo Boost)	4 Cores (HT) x 3,8 GHz (4,2 GHz Turbo Boost)	4 Cores (HT) x 3,8 GHz (4,2 GHz Turbo Boost)
12 GB RAM (DDR3 ECC)	16 GB RAM (DDR4 ECC)	32 GB RAM (DDR4 ECC)	64 GB RAM (DDR4 ECC)
1.000 GB Festplatte (Hardware RAID 1)	1.000 GB Festplatte (Software RAID 1)	2.000 GB Festplatte (Hardware RAID 1)	2.000 GB Festplatte (Hardware RAID 1)
Unlimited traffic	Unlimited traffic	Unlimited traffic	Unlimited traffic
Kostenlose Domain ⓘ	Kostenlose Domain ⓘ	Kostenlose Domain ⓘ	Kostenlose Domain ⓘ
Site Scan ⓘ	Site Scan ⓘ	Site Scan ⓘ	Site Scan ⓘ
Content Delivery Network (CDN) mit Railgun ⓘ	Content Delivery Network (CDN) mit Railgun ⓘ	Content Delivery Network (CDN) mit Railgun ⓘ	Content Delivery Network (CDN) mit Railgun ⓘ

Hosting analysis (Ionos) – Dedicated #2

L-16 SSD Erweiterte Hardware für anspruchsvolle Webprojekte	XL-32 SSD Leistungsstarke Hardware für umfangreiche Webprojekte	XL-64 SSD Leistungsstarke Hardware für mehrere umfangreiche Webprojekte
3 Monate 50,- €/Mon. danach 70,- €/Mon.	3 Monate 80,- €/Mon. danach 100,- €/Mon.	3 Monate 100,- €/Mon. danach 120,- €/Mon.
Weiter	Weiter	Weiter
Intel® Xeon® E3-1230 v6	Intel® Xeon® E3-1270 v6	Intel® Xeon® E3-1270 v6
4 Cores (HT) x 3,5 GHz (3,9 GHz Turbo Boost)	4 Cores (HT) x 3,8 GHz (4,2 GHz Turbo Boost)	4 Cores (HT) x 3,8 GHz (4,2 GHz Turbo Boost)
16 GB RAM (DDR4 ECC)	32 GB RAM (DDR4 ECC)	64 GB RAM (DDR4 ECC)
480 GB Intel® SSD Festplatte (Software RAID 1)	800 GB Intel® SSD Festplatte (Hardware RAID 1)	800 GB Intel® SSD Festplatte (Hardware RAID 1)
Unlimited traffic	Unlimited traffic	Unlimited traffic
Kostenlose Domain ⓘ	Kostenlose Domain ⓘ	Kostenlose Domain ⓘ
Site Scan ⓘ	Site Scan ⓘ	Site Scan ⓘ
Content Delivery Network (CDN) mit Railgun ⓘ	Content Delivery Network (CDN) mit Railgun ⓘ	Content Delivery Network (CDN) mit Railgun ⓘ

Hosting analysis (Ionos) – VPS

	CPU	RAM	SSD	Preis
VPS S	1 vCore	512 MB	10 GB	6 Monate 1,- €/Mon. danach 2,- €/Mon. Konfigurieren
VPS M	2 vCores	2 GB	80 GB	6 Monate 2,- €/Mon. danach 5,- €/Mon. Konfigurieren
VPS L	2 vCores	4 GB	120 GB	6 Monate 5,- €/Mon. danach 10,- €/Mon. Konfigurieren
VPS XL	4 vCores	8 GB	160 GB	6 Monate 1,- €/Mon. danach 16,- €/Mon. Konfigurieren
VPS XXL	6 vCores	12 GB	240 GB	6 Monate 12,- €/Mon. danach 24,- €/Mon. Konfigurieren

Domain analysis (Ionos)

 0,08 €/Mon. im 1. Jahr Prüfen	 0,70 €/Mon. im 1. Jahr Prüfen	 0,10 €/Mon. im 1. Jahr Prüfen	 0,08 €/Mon. im 1. Jahr Prüfen
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Domains - Basics

- Domain name identifies a realm of administrative autonomy, authority or control within the Internet
- Domain name is changed onto numeric address used by the network
- It is an identifier used by clients to get access to the Internet resources (including websites)

Domains - Levels

- Top-Level Domain (TLDs) – e.g. .com
 - Generic Top-Level Domain (gTLD) - .com, .org, .info and .net (original/core ones) + 1200+ others (now)
 - Generic-restricted Top-Level Domain (grTLD) - .gov, .edu or .mil
 - Sponsored Top-Level Domain (sTLD) - .aero (by Société Internationale de Télécommunications Aéronautiques)
 - Country Code Top-Level Domain (ccTLD) – e.g. .pl, .uk, .de or .us (308 ccTLDs existing now)
- Second-Level Domain (SLD) – e.g. XYZ.com
- Third-Level Domain (TLD) – e.g. ABC.XYZ.com

Web-server management #1

Server Setup and Configuration

- **Installation:** Setting up the web server software (e.g., Apache, Nginx, Microsoft IIS) on the hardware or virtual machine.
- **Configuration:** Adjusting server settings to optimize performance, security, and functionality, including setting up virtual hosts, SSL certificates, and directory permissions.

Performance Monitoring

- **Resource Monitoring:** Keeping track of server resources such as CPU usage, memory consumption, disk space, and network bandwidth to ensure optimal performance.
- **Log Management:** Analysing server logs (access logs, error logs) to identify issues, monitor traffic patterns, and troubleshoot problems.

Security Management

- **Access Control:** Implementing user authentication and authorization to restrict access to server resources.
- **Firewalls and Security Protocols:** Configuring firewalls and using security protocols to protect data and prevent unauthorized access.
- **Regular Updates:** Keeping server software and applications up to date to protect against vulnerabilities and security threats.

Backup and Recovery

- **Data Backup:** Regularly backing up website data, databases, and server configurations to prevent data loss.
- **Disaster Recovery:** Establishing a recovery plan to restore services quickly in case of hardware failure, data corruption, or other disasters.

Web-server management #2

Load Balancing and Scalability

- Load Balancing: Distributing incoming traffic across multiple servers to ensure no single server becomes overwhelmed, improving performance and reliability.
- Scalability: Planning for future growth by ensuring the server infrastructure can handle increased traffic and resource demands.

Content Delivery

- CDN Integration: Utilizing Content Delivery Networks (CDNs) to cache and deliver content from geographically distributed servers, improving load times and reducing latency for users.

Application Management

- Deployment: Managing the deployment of web applications, including updates and patches.
- Monitoring Application Performance: Using tools to monitor the performance of web applications and troubleshoot issues.

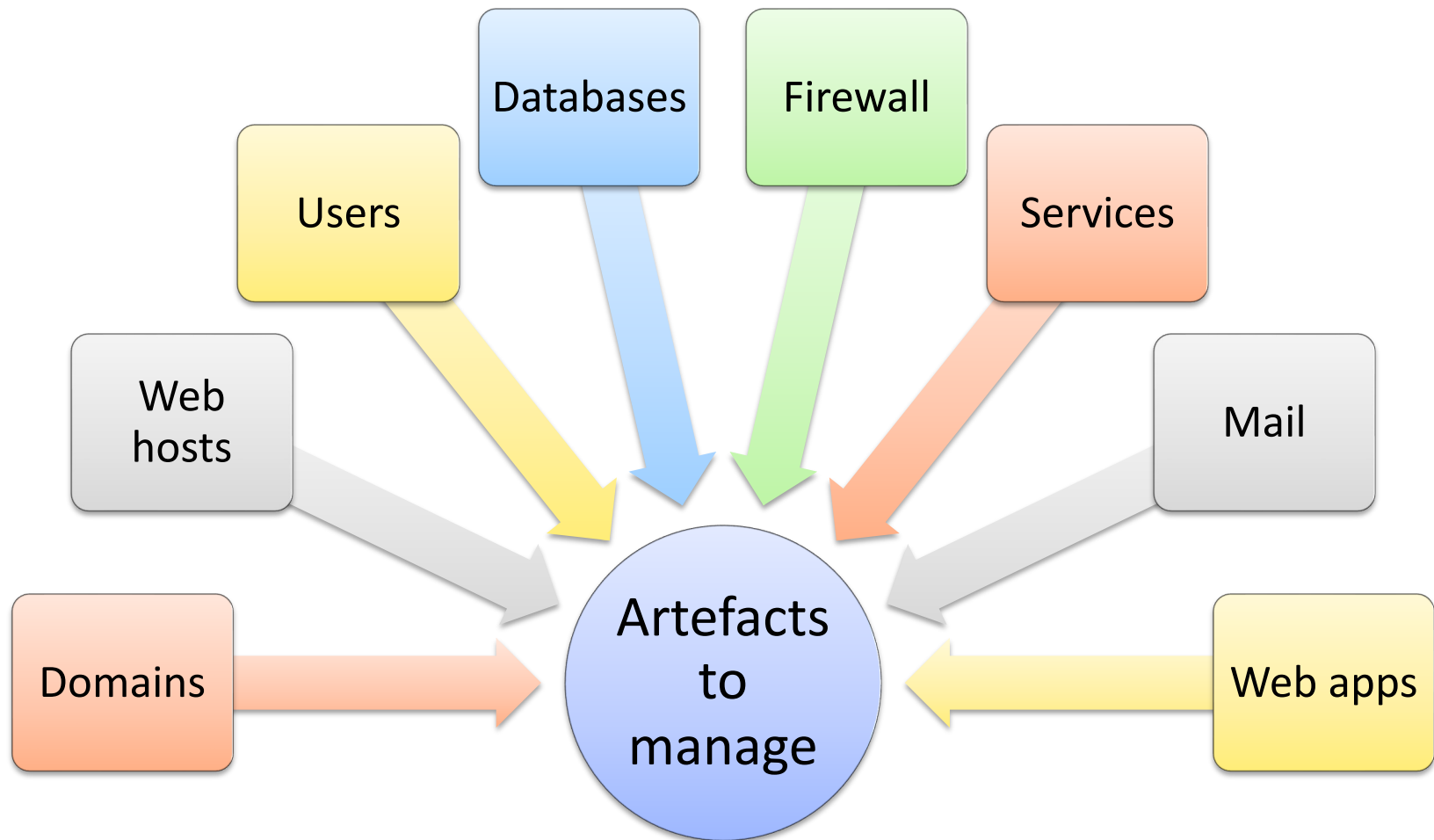
User Support and Troubleshooting

- Technical Support: Providing assistance to users and clients regarding server-related issues.
- Troubleshooting: Diagnosing and resolving server problems, such as downtime, slow performance, or application errors.

Documentation and Reporting

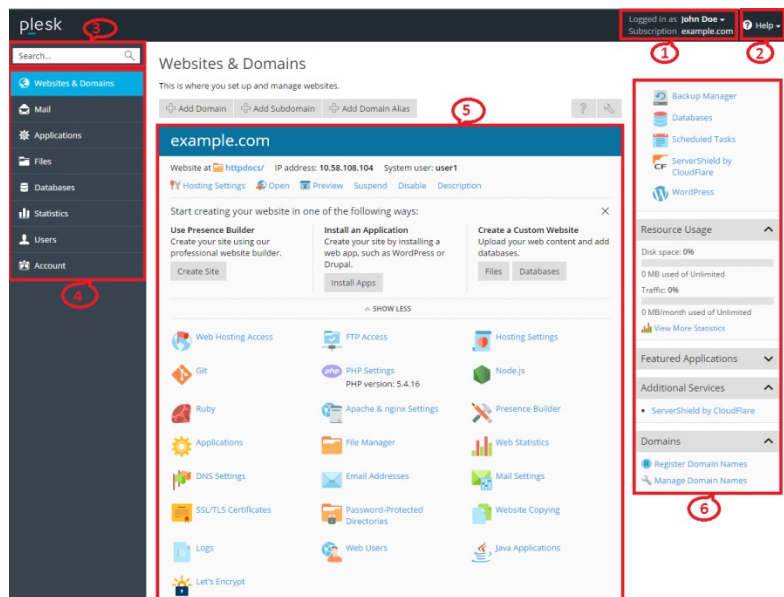
- Documentation: Maintaining detailed records of server configurations, changes, and procedures for future reference.
- Reporting: Generating reports on server performance, security incidents, and resource usage for analysis and decision-making.

Server management software

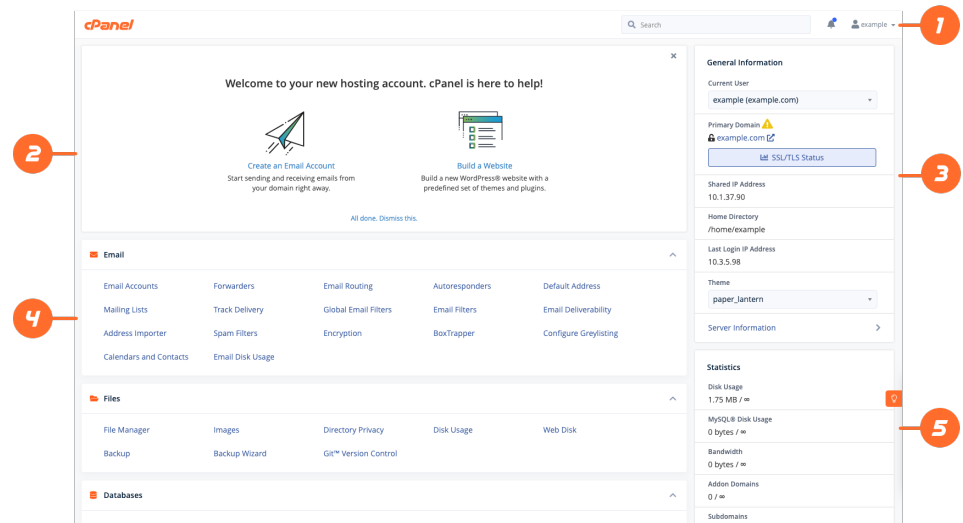


Server management software - Examples

Plesk



cPanel



Full comparison: https://en.wikipedia.org/wiki/Comparison_of_web_hosting_control_panels
Note: These packages are laid over OS commands & services. Expect OS config to be generated on a basis of information stored in database. While using such systems, periodic management at low level might be difficult/impossible (config will be overwritten by higher level software).

Summary

- E-shops have common dependencies, and usually are based on LAMP/WAMP platforms
- Database is an essential piece of every web system
- Internet connection is a gateway connecting system with its users
- E-commerce site must be hosted somewhere and to be accessible (domain)

TILL NEXT LECTURE